

2021 Ph H2 Q8

Section: Particles and Waves

Topic: Photoelectric Effect

Photoelectric experiment with white and ultraviolet light. (a) Explain why UV produces current but white light does not. (b) $E_{\text{photon}} = 8.0 \times 10^{-19} \text{ J}$, $\phi = 6.9 \times 10^{-19} \text{ J}$, supply 12.0 V: (i) calculate KE_{max} , (ii) show kinetic energy gained from acceleration is $1.92 \times 10^{-18} \text{ J}$, (iii) find maximum electron speed.

Worked solution

(a) White light photons have energy below the work function (ϕ), so no electrons are emitted. Ultraviolet photons have energy above ϕ , so electrons are ejected, giving a current.

(b)(i) Maximum KE

$$\begin{aligned} KE_{\text{max}} &= E_{\text{photon}} - \phi \\ &= 8.0 \times 10^{-19} - 6.9 \times 10^{-19} = 1.10 \times 10^{-19} \text{ J} \end{aligned}$$

Answer: $1.1 \times 10^{-19} \text{ J}$

(b)(ii) KE from acceleration

$$KE = eV = (1.60 \times 10^{-19})(12.0) = 1.92 \times 10^{-18} \text{ J}.$$

Shown as required

(b)(iii) Maximum speed

$$\begin{aligned}\text{Total KE} &= \text{KE}_{\text{max}} + eV \\ &= 1.10\text{e-}19 + 1.92\text{e-}18 = 2.03\text{e-}18 \text{ J}\end{aligned}$$

$$\begin{aligned}v &= \sqrt{2E/m} \\ &= \sqrt{(2 \times 2.03\text{e-}18 / 9.11 \times 10^{-31})} = 2.11\text{e+}06 \text{ m/s}\end{aligned}$$

Answer: $2.1 \times 10^6 \text{ m/s}$

Final answers

(a) UV photons exceed work function, white light does not

(b)(i) $\text{KE}_{\text{max}} = 1.1 \times 10^{-19} \text{ J}$

(b)(ii) KE from acceleration = $1.92 \times 10^{-18} \text{ J}$

(b)(iii) $v \approx 2.1 \times 10^6 \text{ m/s}$

Revision tips

- Photoelectric effect: photon must have $E \geq \phi$ to eject electrons.
- $\text{KE}_{\text{max}} = hf - \phi$.
- Electrons accelerated by pd gain extra $\text{KE} = eV$.
- Use $\text{KE} = \frac{1}{2}mv^2$ to find electron speed.
- White light photons (visible) have energies $<$ work function of many metals.